The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 May 1994

INCH-POUND

MIL-S-19500/240F
21 February 1994
SUPERSEDING
~~MIL-S-19500/240E~~
23 September 1982

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER
TYPES 1N645-1, 1N647-1,1N649-1, 1N645UR-1,
1N647UR-1, 1N649UR-1, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for general purpose silicon diode. Three levels of product assurance are provided for each device type as specified in MIL-S-19500 (see 6.3.2).

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Type 1/	V_{RSM}	V_{RWM}	I_O T_A = 150°C	I_O 2/ T_A = 25°C	I_{FSM} t_p = 1/120 s T_A = 25°C	T_{op}	T_{STG}	Barometric pressure, reduced
	V (pk)	V (pk)	mA	mA	A	°C	°C/W	mm Hg
1N645-1	270	225	150	400	5	-65 to +175	-65 to +175	8
1N647-1	480	400	150	400	5	-65 to +175	-65 to +175	8
1N649-1	720	600	150	400	5	-65 to +175	-65 to +175	8

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

2/ Derate 2.0 mA/°C between 25°C and 150°C. Derate 6 mA/°C between 150°C and 175°C.

1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

Type 1/	V_F at I_F = 400 mA dc, 2% duty cycle, 8.3 ms max pulse width	I_R at T_A = 25°C		I_R at T_A = 150°C	
	V dc (max)	µA dc (max) at V_R		µA dc (max) at V_R	
1N645-1	1.0	.050	225 V dc	25	225 V dc
1N647-1	1.0	.050	400 V dc	25	400 V dc
1N649-1	1.0	.050	600 V dc	25	600 V dc

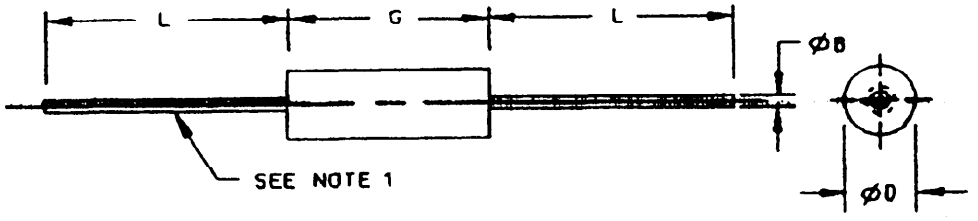
1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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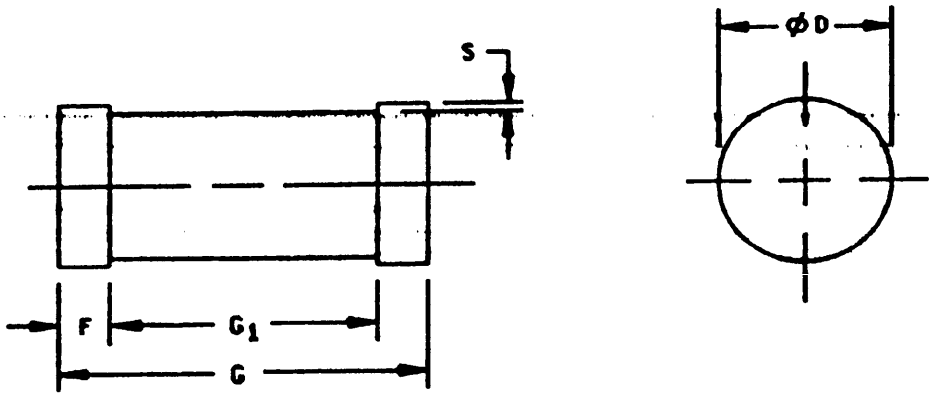


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	0.018	0.022	0.46	0.56	1
ϕD	0.055	0.090	1.40	2.29	
G	0.120	0.200	3.05	5.08	
L	1.000	1.500	25.40	38.10	

NOTES:
1. The specified lead diameters apply in the zone between .050(1.27 mm) from the diode body to the end of the lead. Outside of this zone the lead diameter shall not exceed the maximum dimension ϕB.
2. Metric equivalents are given for general information only.

FIGURE 1. Physical dimensions.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
ϕD	.063	.067	1.60	1.70
F	.016	.022	0.41	0.56
G	.130	.146	3.30	3.71
G_1	.100 ref		2.54 ref	
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions 1N645UR-1, 1N647UR-1, AND 1N649UR-1.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 and 2 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

3.3.2 Dash one construction. Dash one (-1) devices shall be metallurgically bonded double plug construction in accordance with the requirements of category I, II, or III (see MIL-S-19500, paragraph 30.14 and subparagraphs).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Polarity. The polarity shall be indicated with a contrasting color band to denote the cathode end. No color coding will be permitted.

3.4.2 Marking of UR version devices. For UR version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANTXV and JANTX levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)
9	Not applicable
11	I_{R1}, V_{F1}
12	See 4.3.1, t = 48 hours
13 1/	Subgroup 2 of table I herein; $\Delta I_{R1} \leq 100\%$ of initial reading or 10 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.025$ V dc change from initial reading.

1/ Thermal impedance need not be repeated at this step.

4.3.1 Burn-in test. All devices shall be operated under one of the following conditions:

Type 1/	MIL-STD-750, method 1038, condition B (a), T_A = 125°C; f = 60 Hz, (see 4.5.1)	MIL-STD-750, method 1038, condition B (a), T_A = 25 °C; f = 60 Hz, (see 4.5.1)	MIL-STD-750, method 1038, condition B (b), T_A = 125°C
1N645-1	V_R = 225 v(pk); I_O = 200 mA	V_R = 225 v(pk); I_O = 400 mA	I_F = 200 mA
1N647-1	V_R = 400 v(pk); I_O = 200 mA	V_R = 400 v(pk); I_O = 400 mA	I_F = 200 mA
1N649-1	V_R = 600 v(pk); I_O = 200 mA	V_R = 600 v(pk); I_O = 400 mA	I_F = 200 mA

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

4.3.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measurement current- - - - - - - - - - - - 1 mA - 10 mA
b. I_H forward heating current- - - - - - - - - - .5 A - 1.0 A
c. t_H heating time - - - - - - - - - - - - - - - 10 ms
d. t_{MD} measurement delay time - - - - - - - - - 100 us (max)

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

4.4.2.1 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	I_O = 150 mA; f = 50 - 60 Hz, T_A = 150°C (see 4.5.1). V_R = 225 V(pk) for 1N645-1; V_R = 400 V(pk) for 1N647-1; V_R = 600 V(pk) for 1N649-1.
B5	3101 or 4081	L = .375", $R_{\theta JL}$ = 250°C/W; L = 0", $R_{\theta JC}$ = 100°C/W (surface mount); No moving air permitted.
B6	1032	T_A = 175°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	Tension: Test condition A; weight = 2 pounds; t = 30 ±3s. Lead fatigue: Test condition E.(Terminal strength not applicable to surface mount devices.)
C6	1026	I_O = 150 mA;f = 50 - 60 Hz, T_A = 150°C (see 4.5.1) V_R = 225 V(pk) for 1N645-1; V_R = 400 V(pk) for 1N647-1; V_R = 600 V(pk) for 1N649-1.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table I, group A, subgroup 2 herein.

4.4.4.1 Group E inspection, table VII of MIL-S-19500.

Subgroup	Method	Condition	Sampling plan
E1, E2, E3	----	Not applicable	
E4	3101 or 4081	L = .375"; $R_{\theta JL}$ = 250°C/W for non surface mount; L = 0", $R_{\theta JEC}$ = 100°C/W for surface mount. No moving air permitted.	10 devices, c = 0
E5	1001		10 devices, c = 0

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Steady-state operation life. A half-sine wave of the specified peak voltage shall be impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180 degrees nor less than 150 degrees.

4.5.2 Pulse measurements. Conditions of pulse measurements shall be specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		35	°C/W
Forward voltage	4011	I_F = 400 mA dc (Pulsed, see 4.5.2); 2% duty cycle maximum, 8.3 ms maximum pulse width.	V_{F1}	.8	1.0	V dc
Reverse current	4016	DC method	I_{R1}			
1N645-1		V_R = 225 V dc			0.05	µA dc
1N647-1		V_R = 400 V dc			0.05	µA dc
1N649-1		V_R = 600 V dc			0.05	µA dc
Subgroup 3						
High temperature operation		T_A = 150°C				
Forward voltage	4011	I_F = 400 mA dc (Pulsed, see 4.5.2); 2% duty cycle maximum, 8.3 ms maximum pulse width.	V_{F2}	.7	.95	V dc
Reverse current	4016	DC method	I_{R2}			
1N645-1		V_R = 225 V dc			25	µA dc
1N647-1		V_R = 400 V dc			25	µA dc
1N649-1		V_R = 600 V dc			25	µA dc
Low temperature operation		T_A = -55°C				
Forward voltage	4011	I_F = 400 mA dc (Pulsed, see 4.5.2); 2% duty cycle max, 8.3 ms max pulse width.	V_{F3}		1.2	V dc
Subgroup 4						
Reverse current at peak reverse voltage	4016	AC method; f = 60 Hz	I_{RM1}			
1N645-1		V_R = 270 V(pk)			50	µA(pk)
1N647-1		V_R = 480 V(pk)			50	µA(pk)
1N649-1		V_R = 720 V(pk)			50	µA(pk)
Capacitance	4001	V_R = 4 V dc; f = 1 MHz; V_{sig} = 50 mV(pk-pk) maximum.	C		20	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	Mounting conditions in accordance with MIL-STD-750, test method 1026 T_A = 25°C, I_{FSM} = 5 A; I_O = 150 mA ten 1/120 s surges; 1 surge per minute.				
Electrical measurements		See table II, steps 1, 3, and 5.			75	µA dc
Subgroup 7						
Reverse current, average	4046	T_A = 150°C; I_O = 150 mA	I_R (AV)			
1N645-1		V_R = 225 V(pk)			100	µA
1N647-1		V_R = 400 V(pk)			100	µA
1N649-1		V_R = 600 V(pk)			100	µA

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

TABLE II. Groups A, B, and C electrical measurements. 2/, 3/, 4/

Step	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	Pulse width = 8.3 ms maximum; duty cycle = 2% maximum; I_F = 400 mA dc (pulsed)	V_{F1}	.8	1.0	V dc
2.	Forward voltage	4011	Pulse width = 8.3 ms maximum; duty cycle = 2% maximum; I_F = 400 mA dc (pulsed)	V_{F2}	.8	1.1	V dc
3.	Reverse current	4016	DC method;	I_{R1}			
	1N645-1		V_R = 225 V dc			.050	µA dc
	1N647-1		V_R = 400 V dc			.050	µA dc
	1N649-1		V_R = 600 V dc			.050	µA dc

See footnotes at end of table.

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TABLE II. Groups A, B, and C electrical measurements - continued. 2/, 3/, 4/

Step	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
4.	Reverse current	4016	DC method;	I_{R3}			
	1N645-1		V_R = 225 V dc			.050	µA dc
	1N647-1		V_R = 400 V dc			.050	µA dc
	1N649-1		V_R = 600 V dc			.10	µA dc
5.	Reverse current	4016	DC method; T_A = 150°C	I_{R2}			
	1N645-1		V_R = 225 V dc			25	µA dc
	1N647-1		V_R = 400 V dc			25	µA dc
	1N649-1		V_R = 600 V dc			25	µA dc
6.	Reverse current at peak reverse voltage	4016	AC method;	I_{RM2}			
	1N645-1		V_R = 270 V(pk)			200	µA(pk)
	1N647-1		V_R = 480 V(pk)			200	µA(pk)
	1N649-1		V_R = 720 V(pk)			200	µA(pk)
7.	Reverse current	4016	DC method;	ΔI_{R1} 2/			
	1N645-1		V_R = 225 V dc			100% of initial value or ±.010 µA dc, whichever is greater	
	1N647-1		V_R = 400 V dc				
	1N649-1		V_R = 600 V dc				

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

2/ Devices exceeding group A limits for this test shall not be acceptable.

3/ The electrical measurements for table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, and 5.

b. Subgroups 3 and 6, see table II herein, steps 2, 4, 5, 6, and 7.

4/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1, 3, and 5 (for JAN, JANTX, and JANTXV.

b. Subgroup 6, see table II herein, steps 2, 4, 5, 6, and 7 (for JAN, JANTX, and JANTXV).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish (see 3.3.1).

6.2.1 Substitutability. Non dash-one devices have been deleted from this specification. Dash-one devices are a direct substitute for non dash-one devices and are preferred. Devices in stock are acceptable provided the date code does not exceed 10 February 1993. Existing supplies of the non dash-one parts may be used until exhausted.

6.2.2 Cross reference substitution list. JANS level will no longer be built to this specification after the dating of MIL-S-19500/587. Devices required for space flight applications are found in MIL-S-19500/587. JANS devices built to MIL-S-19500/240 in stock are acceptable provided the date code does not exceed the date of implementation of this specification. A PIN for PIN replacement table follows, and these devices are directly interchangeable:

Non-preferred PIN	Preferred PIN
1N645-1	1N6661
1N647-1	1N6662
1N649-1	1N6663

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99
NASA - NA

Preparing activity:
DLA - ES

(Project 5961-1536)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/240F	2. DOCUMENT DATE (YYMMDD) 94/02/21

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER TYPES 1N645-1, 1N647-1,1N649-1, 1N645-1UR, 1N647-1UR, 1N649-1UR, JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048 (2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) DESC-ECT Dayton, OH 45444-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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